

AI Job Agent

Cleanup, Pricing & Transition Plan

Analysis-Only Decision Checkpoint — Provider-First Clean Architecture

Branch: feat/provider-first-clean-architecture

HEAD: b41454a (includes e9ba070, b41454a)

Prepared: 2026-09-26 · Revised: 2026-09-26 (post-decision correction)

Status: No implementation performed. Analysis only.

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Legend: **FACT** verified directly from repo/DB/live n8n · **INFERENCE** reasoned conclusion from facts · **RECOMMENDATION** my proposal, not yet decided · **FOUNDER DECISION** requires your explicit approval before acting · **SUPERSEDED** a recommendation this report made that the founder has since explicitly declined.

Revision note (2026-09-26): this report originally recommended two quota changes (Student revisions 1→2, Pro matches 45→50) and classified the Job Ingestion Pilot Orchestrator as becoming the production Tier A runner. Both were superseded by explicit founder decisions later in the same session; pricing/quotas stay exactly as originally shipped, and the Pilot Orchestrator is kept only as a pattern/reference source, not the production ingestion foundation. Sections below are corrected accordingly; the original recommendations are preserved, struck through context, and marked **SUPERSEDED** rather than deleted, so the decision trail stays honest.

1. Executive Summary

The repository is in good shape: 92 reviewed migrations, RLS enforced everywhere, checkout/pricing already server-authoritative, and three ATS adapters (Greenhouse/Lever/Workable) proven end-to-end with real retry/dedup/idempotency logic. But almost the entire product past "ingest a job" is schema-only — **matches, cover letters, applications, and notifications all have zero rows and zero worker code**. The company-registry research effort (593 `company_sources` rows, dozens of CSVs, multiple Apify pilots across five countries) vastly outpaces what has actually been ingested (20 jobs, all from one bounded local pilot).

The single most urgent operational finding: the live **Source Intelligence Analyzer is still running on a daily schedule against the shared database**, using an older version of its own logic than what this branch just committed. It wrote 10 real rows today with the new `retryable` column left unset — not a bug, just proof the live n8n instance has drifted behind the repository. This needed a founder decision — since resolved: the workflow has been manually deactivated and is now frozen/legacy.

Pricing is well-engineered at the database layer (price versions, locked subscriptions, tamper-proof checkout). This report originally proposed two quota adjustments — Student cover-letter revisions (1→2) and Pro match quota (45→50) — as worth founder sign-off. **SUPERSEDED** : the founder has since explicitly confirmed pricing and all quotas remain exactly as originally shipped (Free \$0 / Student \$9, 25 matches, 8 cover letters, 1 revision / Pro \$18, 45 matches, 15 cover letters, 3 revisions) — see Parts 6, 7, 10, 16, 29. Pro's price itself was correctly left unchanged at launch; the repo's own price-versioning migration rehearses a future \$22 Pro price as an example, evidence to revisit after usage data, not before.

Cover-letter generation is a commercial promise in every plan today — it must be production-ready before paid public launch, even though it is correctly built after ingestion and matching. Recommended cleanup order: freeze legacy, build the new pipeline, archive superseded work only after its replacement is proven — not a cleanup-first pass that risks nothing while the product stands still.

This report also originally classified the Job Ingestion Pilot Orchestrator as becoming the production Tier A runner for the new architecture.

SUPERSEDED : the founder has since confirmed it is **not** the production foundation for the new provider-first ingestion layer — it is preserved only as a reference/pattern source (retry/backoff, deduplication, idempotency, bounded writes, stale-close, error handling). The new ingestion layer will be built cleanly. See Parts 18, 19, 24, 27.

2. Verified Current Branch State

Check	Result
Current branch	feat/provider-first-clean-architecture
HEAD	b41454a
git status	Clean — no staged, unstaged, or untracked changes
HEAD contains	e9ba070 (source-intelligence slice) and b41454a (docs checkpoint) – confirmed
Migrations	92 files, chronologically consistent, no gaps found
Row counts (read-only, local dev DB)	jobs =20, matches =0, cover_letters =0, applications =0, notifications =0, automation_tasks =0, companies =559, company_sources =593, source_intelligence =60, audit_events =513

FACT No files were modified, no migration run, no workflow touched, nothing committed, during this audit. All commands used were read-only (git status/log , select count , search_workflows / get_workflow_details).

3. Architecture Confirmation

FACT Option B (clean layer, same project, same database) is directly supported by repo evidence: RLS, auth, onboarding, profiles, cvs, entitlements, plans/pricing, companies/company_sources, and the three ATS adapters are all mature and worth keeping. Nothing found in this audit argues for a rebuild or a new project.

INFERENCE The company registry's current role (a gate — `automation_eligibility / review_status` deciding whether a source may be ingested at all) is exactly what Option B wants to retire. The schema already supports the target role without any migration: `companies` = identity, `company_sources` = trust/provenance record, and nothing currently in the schema forces company-source approval to happen before a job can be created (`jobs.source_id` is nullable, `admin_manual` jobs already bypass it). The gate is a *workflow behavior* (Job Ingestion Pilot Orchestrator re-checking `review_status` live before writing), not a database constraint — so "stop gating" is achievable by changing workflow logic, not schema.

FACT Registry Sync and Source Intelligence Analyzer both already write to `company_sources / source_intelligence` reactively (trust escalation), not as a precondition for ingestion elsewhere in the schema. Their current *usage* (Source Intelligence Analyzer gating the pilot's "Is Source Approved?" check) is what needs to change, not their tables.

4. Current Pricing Plan Review

FACT The live source of truth (`public.plans` seeded by `20260802090000_create_plans.sql`, priced by `plan_prices / price_versions`) matches exactly what you pasted: Free \$0 (1 match / 1 cover letter), Student \$9/mo (25 matches / 8 cover letters), Pro \$18/mo (45 matches / 15 cover letters). `docs/PRICING.md` confirms no price has changed since launch. **Confirmed by the founder as final and unchanged** — see Part 29.

FACT There is **no revision-limit column anywhere in the schema**. `cover_letters.revision` is a bare integer counter with no plan-aware cap logic. "1 revision (Student) / 3 revisions (Pro)" is currently a marketing promise only — enforcing it is net-new application logic, not something already half-built. These exact numbers (1 / 3) are the founder-confirmed final values to enforce — see Part 10.

FACT Checkout is fully server-authoritative and tamper-tested (`tests/db/checkout-tampering.test.mjs`, `tests/unit/checkout-request-tampering.test.mjs`) — a price/quota change only ever requires one `publish_price_version()` call plus (for quotas) an edit to the `plans` row. No frontend price constant exists to keep in sync.

5. Free Plan Review

Question	Assessment
Is 1 match/month useful enough?	RECOMMENDATION Yes — it is a taste, not a habit, and that is the correct job for Free. Enough to prove the match/explanation quality; not enough to replace paying.
Is 1 cover letter/month appropriate?	Yes — naturally paired 1:1 with the 1 match. No change.
Should tracking be included?	Yes — marginal cost is a handful of DB rows; letting Free users see the tracking dashboard populate is a real part of the upgrade pitch.
Is manual apply only correct?	Yes — zero send-side cost, zero abuse risk, and a clean, unambiguous upgrade lever (email-apply is the paid unlock).

RECOMMENDATION KEEP Free exactly as-is. **FACT** Founder-confirmed final.

6. Student Plan Review

Question	Assessment
Is 25 curated matches/month enough?	Likely generous, not scarce — Lebanon is a genuinely small market. The real risk is supply (can ingestion + matching produce 25 <i>good</i> matches/month for one small country?), not demand. Founder-confirmed final: stays at 25.
Daily search, capped delivered matches?	Yes — this is the correct model and is addressed explicitly in Part 8/11 below.
Are 8 cover letters/month appropriate at \$9?	Yes, roughly 1 letter per 3 matches — a reasonable ratio. Founder-confirmed final: stays at 8.
Is 1 revision per letter appropriate?	SUPERSEDED This report originally argued 1 was too stingy and recommended raising to 2 (see the original Part 10 reasoning, preserved there for the record). The founder has since explicitly confirmed Student stays at 1 revision per letter , unchanged. No further argument from this report.
Is Student sufficiently better than Free?	Yes — 25 vs 1 matches, 8 vs 1 cover letters, and email-apply unlock is a clear, large step up.

7. Pro Plan Review

Question	Assessment
Is 45 matches/month enough?	SUPERSEDED This report originally called 45 borderline low relative to Pro's 5-market coverage and recommended raising to 50. The founder has since explicitly confirmed Pro stays at 45 matches/month , unchanged. No further argument from this report.
Would 50/60 make more sense?	SUPERSEDED — moot; 45 is the founder-confirmed final number.
Are 15 cover letters/month appropriate?	Yes, no change. Founder-confirmed final: stays at 15.
Are 3 revisions per letter too generous?	No — keep at 3. Founder-confirmed final. It remains a clear, cheap differentiator over Student's (unchanged) 1.
Is \$18 sustainable?	Yes with real margin to spare — see Part 14/15.
Is geographic expansion enough to justify upgrade?	Yes, it is the single clearest upgrade story in the product (Lebanon-only vs. relocation to 4 additional countries + verified international remote).

8. Match Quota Recommendation

RECOMMENDATION Definition (must be picked exactly once, server-enforced): a "curated match" consumed against the plan's monthly quota is a distinct `(user_id, job_id)` pair that reaches `matches.status = 'pending_review'` for the first time in the current billing period — i.e., a genuinely new match surfaced to the user, not a discovery/scoring/notification event.

- **Discovered** (job ingested) — never consumes.
- **Scored** (embedding + LLM rerank run) — never consumes by itself; most scored candidates are discarded below threshold and must never reach the user.
- **Visible/notified** — this is the consuming event, and only the first time for that `(user, job)` pair.
- A duplicate job reappearing, or a re-score under the *same* confirmed `cv_analysis_id`, is a no-op against the existing unique row `(matches_user_job_analysis_key)` — structurally cannot consume twice.
- A genuinely new match created because the user confirmed a *new* CV analysis is, by design, a new row — and does consume, because it is a new delivered match, not a recompute.

Quota (founder-confirmed final): Free=1, Student=25, Pro=45. **SUPERSEDED note** : this section originally read "Pro=50 (raised from 45)" — that quota change was declined; Pro's quota is unchanged at 45. This is enforceable with a single monthly count query against `matches` filtered by `created_at` in the current period and `user_id`, at the point a new match row is about to be inserted.

9. Cover Letter Quota Recommendation

RECOMMENDATION A cover-letter unit is consumed only on a **successful** generation that persists content (`generation_status='completed'` with non-null `generated_content`). A failed generation attempt, or a retry caused by our own platform error, must never consume — the current schema already has the right shape for this (`generation_status` enum `pending/completed/failed`); the missing piece is simply gating the quota check on the transition into `completed` , not on the request.

Quotas (founder-confirmed final, unchanged from launch): Free=1, Student=8, Pro=15.

FACT Important schema constraint: `cover_letters_match_id_key` is a **unique index on** `match_id` — there can only ever be *one* cover-letter row per match. "Regenerate from scratch" and "revise" are therefore the same operation at the database layer today (an update to `generated_content` / `edited_content` on the existing row, with `revision` incremented) — there is no way to hold two independent drafts for one match without a schema change. This must inform Part 10's design, not be discovered later.

10. Revision Quota Recommendation

SUPERSEDED This report originally recommended Student 1 → 2 revisions per letter (Pro unchanged at 3). The founder has since explicitly confirmed both values stay exactly as originally shipped: **Student = 1 revision per letter, Pro = up to 3 revisions per letter**. No further argument from this report. The original reasoning is preserved below for the record only.

Original recommendation (declined): "Student: 1 → 2 revisions per letter. Pro: keep 3. A revision consumes 1 unit of that letter's own counter (not the account-wide cover-letter quota) only when it is a genuine user-requested edit that produces new persisted content. A platform-error regeneration (we failed to save, we time out, we crash mid-generation) must replay the same attempt without incrementing anything — tie this to an idempotency key per generation attempt, the same pattern already used for `applications.idempotency_key`."

INFERENCE (still current, unaffected by the quota decision) Because of the one-row-per-match constraint (Part 9), "revision" must be implemented as in-place updates with a monotonically increasing `revision` counter and a cap check (`revision < plan_limit + 1`, using the final limits of 1 for Student and 3 for Pro) before allowing another edit — not as new rows. This is a small, well-scoped addition to `save_cover_letter_edit` (already referenced in the applications migration's comments as the intended entry point), not a new subsystem. A platform-error regeneration must still never consume a revision, regardless of the limit's value.

11. CV Tailoring Scope

FACT Current plan copy (all three tiers) promises AI **cover letters**. None of the three plans promise a rewritten CV per job.

RECOMMENDATION Keep CV tailoring suggestions later/optional — it is not a launch commitment, and building it now would compete for engineering time against cover-letter generation, which is a launch commitment (Part 12 below). Cover-letter generation is paid-launch scope; CV tailoring is a clean, additive post-launch feature with its own future pricing conversation if it ever ships.

12. Application Tracking MVP

FACT The existing `applications.status` check constraint only allows `pending_send / sending / sent / failed / cancelled` — this is the *send-attempt* lifecycle (did our own dispatch succeed), not the human *outcome* lifecycle you listed (`prepared/approved/applied/confirmation_received/interview/rejected/offer/no_response/withdrawn`).

RECOMMENDATION Yes — MVP should support the outcome lifecycle as user-editable manual states, additive to (not replacing) the existing technical `status` column. Concretely: a new small, user-writable field (or a separate `application_outcomes` row keyed by `application_id`) the user sets themselves from the dashboard, defaulting from `applications.status` (e.g. `sent` → default outcome `applied`) and editable by the user thereafter. Gmail-based auto-parsing is correctly deferred — do not let it block launch. This is exactly the kind of small, additive, reviewable migration Part §17 of AGENTS.md already expects.

13. Email Apply Approval Rule

FACT The schema already enforces the described flow structurally, before any send-worker exists: `applications.approved_at` and `approved_by` are **NOT NULL at creation** — a row cannot exist without approval evidence — and `enforce_application_method_matches_job` independently blocks `application_method='email'` for any job whose own `application_method` isn't `'email'` (every LinkedIn listing, by construction).

INFERENCE There is currently zero code path anywhere in the repo that creates an application without going through this gate — no auto-apply, no threshold-triggered send, no cron that inserts an application. The architecture confirmation in your prompt (prepare → review → edit → explicit approve → only then send) is already the only path the schema allows; the future work is purely building the sender on top of an invariant that already exists, never re-deriving it.

14. Cost Model

RECOMMENDATION Ranges below use current-generation embedding + structured-generation LLM pricing as a sanity check against the existing architecture reports, not a blind restatement of them — treat as directional, not exact. Figures below use the founder-confirmed final quotas (Student: 25 matches / 8 letters / 1 revision; Pro: 45 matches / 15 letters / 3 revisions).

Tier	Scenario	Drivers	Est. monthly variable cost	
Free	Any usage (capped at 1+1)	~10–30 jobs embedded/reranked to surface 1 match; 1 cover letter	\$0.05 – \$0.30	
Student	Low	Few matches claimed, 2–3 cover letters, 0–1 revisions	\$0.30 – \$1.00	
		Normal	~15–20 of 25 matches, ~6 cover letters, ~1 revision total across letters	\$1.00 – \$3.00
		Heavy	Full 25 matches, full 8 cover letters, full 1 revision each	\$3.00 – \$5.50
Pro	Low	Few matches, 3–4 cover letters	\$1.00 – \$2.50	
		Normal	~25 of 45 matches, ~10 cover letters, ~1–2 revisions each	\$2.50 – \$5.50
		Heavy	Full 45 matches, full 15 cover letters, full 3 revisions each	\$5.50 – \$11.00

SUPERSEDED note : this table originally sized Student's Heavy scenario around 2 revisions/letter and Pro's around 50 matches, reflecting the two declined quota changes. Figures above are corrected to the founder-confirmed final quotas (Student 1 revision, Pro 45 matches); the cost impact of the correction is minor (a few tens of cents at the Heavy end) and does not change any conclusion below.

INFERENCE Cover-letter generation and revisions dominate variable cost, not match scoring — embeddings and LLM rerank against a shortlist are the cheapest step in the pipeline.

15. Pricing Sustainability

Plan	Revenue	Est. cost range	Est. gross margin
Free	\$0	\$0.05–\$0.30	Negative — a deliberate acquisition cost, not a margin line
Student	\$9.00	\$0.30–\$5.50	~39%–97%
Pro	\$18.00	\$1.00–\$11.00	~39%–94%

INFERENCE Even at the heavy-usage tail, both paid plans stay solidly margin-positive at the founder-confirmed final quotas. Cost is not the binding constraint on today's price points — the real constraint on growth is **supply** (verified job sources per market), which no price or quota change fixes.

16. Recommended Final Feature Matrix

SUPERSEDED This section originally proposed two changes from the current UI (Student revisions 1→2, Pro matches 45→50), both marked **FOUNDER DECISION** and pending. The founder has since explicitly declined both. The matrix below reflects the actual final, unchanged plan — there are now **zero** changes from the current UI.

	Free	Student	Pro
Price	\$0	\$9/mo	\$18/mo
Geography	Lebanon	Lebanon	Lebanon + Saudi/Qatar/Kuwait/UAE + verified intl remote
Ingestion/search cadence	Daily (system-side)	Daily (system-side)	Daily (system-side)
Curated matches/month	1	25	45 (unchanged)
Match explanation + missing skills	Basic	Full	Full
Cover letters/month	1	8	15
Revisions/letter	—	1 (unchanged)	3
CV tailoring	Not offered (post-launch, all tiers)	Not offered	Not offered
Application tracking	Yes (manual outcome states)	Yes	Yes
Email-apply	No	Yes, approval-gated	Yes, approval-gated
Relocation support	No	No	Yes (explicit preference)
Notifications	Yes	Yes	Yes

Zero changes from current UI — founder-confirmed final. The two cells above are marked only to show they were reconsidered and explicitly kept as-is, not because anything changed.

17. Repo Legacy Audit

A. Runtime code

Frontend (`src/app` , `src/components`) and most of `src/lib` (auth, entitlements, plans, payments, cvs, cvAnalysis, jobPreferences, locations, countries, universities, majors, targetRoles, dailyNews) are current, exercised by tests, and match Option B's "preserve" list exactly. No dead runtime code found in `src/`.

B. Database migrations/schema

All 92 migrations are additive history — Postgres migrations are never deleted in this repo's own convention (AGENTS.md §17/§20), and none should be now. `matches` / `cover_letters` / `applications` / `notifications` are explicitly documented in their own migration comments as "Foundation only — no worker built here" — this is not legacy, it is the exact schema Part 21's matching architecture will build on.

C. Tests

66 test files, all passing as of the last full run this session (workflow: 299/299, db slice: 94/94). No orphaned test file found referencing removed code.

D. Source/company CSV data

`docs/job-source-discovery/*.csv` (8 country/region files) plus five `pilots/` subdirectories (apify-lebanon, apify-uae, himalayas, kuwait-expansion, qatar-expansion, saudi-expansion) totalling 80+ CSVs. These are real, valuable research — the 593-row `company_sources` table was imported from exactly this data via `scripts/import-company-registry.mjs`. Not useless, but also far ahead of what ingestion can currently consume.

E. n8n workflow definitions

See Part 19 for full reconciliation. Five workflows tracked in `n8n-workflows/`; three untracked/shadow workflows found live, created today.

F. Research docs/reports

`docs/*.md` and the four PDFs already checkpointed (`docs/*.pdf`) are the architecture-decision record this very plan builds on. Archival-important, not runtime.

G. Temporary scripts

`scripts/` (5 files) are all either test infrastructure (`run-db-tests.mjs` , crash-recovery sweep) or a real, idempotent, re-runnable import script (`import-company-registry.mjs`). None are throwaway.

H. Generated artifacts

`.next/` , `tsconfig.tsbuildinfo` — build output, already gitignored/regenerable, not part of this audit's scope.

I. Historical architecture experiments

Top-level `AI_JOB_AGENT_SYSTEM_AUDIT.md` , `AI_JOB_AGENT_FOUNDATIONS_IMPLEMENTATION_REPORT.md` , `DATABASE_PLAN.md` , `daily-news-migrations-report.md` sit at repo root rather than under `docs/` — historical, still occasionally referenced by migration comments (see companies migration citing the readiness audit), so not dead, just misplaced.

18. File-by-File Cleanup Classification

Path	What it does	Imported by runtime?	Tests reference?	Classification	Replaced by
n8n-workflows/source-intelligence-analyzer.ts/.json	Classifies unknown sources; retry/promotion logic just added	N/A (n8n)	Yes, 2 files	KEEP BUT CHANGE ROLE	Stays reactive trust-escalation, exits critical path
n8n-workflows/registry-sync.ts/.json	Converges discovery input into <code>resolve_registry_candidate()</code>	N/A	Yes, 2 files	KEEP — STILL USED BY NEW ARCHITECTURE	—
n8n-workflows/job-ingestion-pilot-orchestrator.ts/.json	Proven Greenhouse/Lever/Workable ingestion, inactive/manual	N/A	Yes	FREEZE — PATTERN/REFERENCE SOURCE ONLY SUPERSEDED classification	Its retry/backoff, dedup, idempotency, bounded-write, and stale-close patterns are extracted into the new clean provider-first ingestion layer. The workflow itself is not the production Tier A runner — founder-confirmed. (Originally classified here as "KEEP — STILL USED BY NEW ARCHITECTURE / becomes Tier A of the new pipeline"; superseded.)
n8n-workflows/cv-analysis-worker.ts/.json	CV parsing worker; live as "CV Analysis Worker v2" with node-name drift	N/A	Yes	REVIEW BEFORE ARCHIVE (drift, not deletion)	Reconcile naming, keep
n8n-workflows/error-handler.ts/.json	Reusable failure handler for scheduled workflows	N/A	Yes	KEEP AS-IS	—
docs/job-source-discovery/*.csv (8 files)	Country/region company research, imported into <code>company_sources</code>	Yes, via import script (already run)	Indirectly (via seeded DB rows)	KEEP — historical seed/trust data	—
docs/job-source-discovery/pilots/apify-lebanon/**	Apify company-discovery pilot raw runs + enrichment	No	No	ARCHIVE (evidence, not runtime)	Superseded by promoted registry rows
docs/job-source-discovery/pilots/apify-uae/**	Same, for UAE	No	No	ARCHIVE	Same
docs/job-source-discovery/pilots/kuwait-expansion/**, qatar-expansion/**, saudi-expansion/**	Manual/Apify country-expansion research	No	No	ARCHIVE	Same
docs/job-source-discovery/pilots/himalayas/**	Evaluated a real international-remote board — Tier D candidate	No	No	REVIEW BEFORE ARCHIVE — worth revisiting as Tier D evidence	Possible future adapter
docs/job-source-discovery/source-expansion/**	Apify discovery design + source catalog notes	No	No	ARCHIVE	—
scripts/import-company-registry.mjs	Idempotent CSV → <code>companies</code> / <code>company_sources</code>	No (invoked)	Indirectly	KEEP AS-IS	—

	importer	manually)			
docs/job-ingestion-database-readiness-audit.md, job-ingestion-pilot.md, n8n-workflow-change-process.md, PRODUCTION_READINESS.md, PRICING.md	Living operational/architecture docs	N/A	Cited by migration comments	KEEP AS-IS — MUST NOT DELETE	—
AI_JOB_AGENT_SYSTEM_AUDIT.md, AI_JOB_AGENT_FOUNDATIONS_IMPLEMENTATION_REPORT.md, DATABASE_PLAN.md, daily-news-migrations-report.md (repo root)	Early-project audits/plans	N/A	Cited by migration comments	KEEP BUT CHANGE ROLE — move under docs/archive/ location only, content untouched	—
docs/*.pdf (4 files just checkpointed)	Architecture-decision record behind Option B	N/A	No	MUST NOT DELETE	—

FACT Nothing in this table has zero dependents and zero historical value — there is no "safe to delete now" candidate in this repository today. Everything old is either still referenced, still seed data, or still evidence for a decision already made.

19. n8n Workflow Reconciliation

Workflow	Repo?	Live?	Active/Sched.	Drift	Classification
Source Intelligence Analyzer	Yes (just updated w/ retry+promotion)	Yes, id yM78i3aqFy8DDkPf	Inactive — manually deactivated by the founder; independently confirmed via search_workflows	Live had drifted behind repo before deactivation (only 1 occurrence of "retryable" in its whole definition vs. many in repo)	FREEZE / LEGACY — confirmed correct, no further action required
Registry Sync	Yes	Yes, id LpDt8c8811kTCF2a	Inactive, manual trigger only	Live updatedAt 2026-09-24T15:05 vs repo file 18:26 same day — repo ahead, low risk (inactive)	KEEP — STILL USED BY NEW ARCHITECTURE
Job Ingestion — Pilot Orchestrator	Yes	Yes, id 35s0xBcRRazBX3T1	Inactive, no schedule	Live updatedAt 2026-09-16 vs repo 2026-09-20 — repo ahead, low risk (inactive)	FREEZE — PATTERN/REFERENCE ONLY, NOT THE PRODUCTION FOUNDATION SUPERSEDED classification (originally read "KEEP — becomes Tier A of new pipeline"; the founder has since confirmed the new provider-first ingestion layer is built cleanly, reusing only its proven retry/backoff/dedup/idempotency/bounded-write/stale-close/error-handling patterns, not the workflow itself)
CV Analysis Worker (repo) / "CV Analysis Worker v2" (live)	Yes	Yes, id x5FEh3tUshX15D1r	Active, polling schedule ("Poll Queue Every 5s")	Live node names carry a "1" suffix (e.g. Po11 Queue Every 5s1) not present in repo — cosmetic naming drift from a past duplicate/import, functional equivalence not yet re-verified	REVIEW BEFORE further changes — diff and confirm parity before touching either side. Untouched per founder instruction.
AI Job Agent - Error Handler	Yes	Yes, id GRkqmVu5eWZdKRYU	Inactive (attached manually per test comment)	Minor — live slightly behind repo, inactive so low risk	KEEP AS-IS — preserved as generic infrastructure
"AI agent that can scrape webpages"	No	Yes, id hh0FcU72s8Z2pbRs , created today	Inactive	Not in repo at all — shadow/untracked	EXPERIMENTAL / SHADOW — do not build on; classify only
"Enrich LinkedIn profiles in Google Sheets with Apify"	No	Yes, id 0xDiD4afwnFcjNxf , created today	Inactive	Shadow, LinkedIn-related	EXPERIMENTAL / SHADOW — never activate or build on, per hard product rule against LinkedIn scraping
"Generate leads with Google Maps"	No	Yes, id h62kjGd6MHc009sh , created today	Inactive	Shadow, unrelated to job ingestion as scoped	EXPERIMENTAL / SHADOW — classify only, no engineering time
"Lebanon Tech Hiring Intelligence"	No	Yes, id OAAG2Z7NS6k5UtNq , created 2026-09-02	Inactive	Shadow — predates current architecture, no description available without enabling MCP access (not done, would modify workflow settings)	REVIEW BEFORE ARCHIVE — founder should confirm purpose
"AI/TECH NEWS AUTOMATION"	No	Yes, id KndfnJB3uTSsxk	Active — founder-confirmed: intentionally active, powers the website's AI/Tech News section	Not in n8n-workflows/ — an unversioned production feature, per the founder	DO NOT TOUCH — confirmed in-scope and untouched, per explicit founder instruction
Everything else returned by the search (KAN-113, Personal)	No	Yes/mixed	Mixed	Clearly unrelated personal/other-project workflows	OUT OF SCOPE — not evaluated further

Assistant – Document Ingestion, Agentic Personal Assistant – Chat, Traditional RAG, XAU Gold Price Alert)				sharing the same n8n instance	
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MCP access is disabled for the three shadow experiments and one ambiguous workflow, so their internal nodes/credentials could not be inspected without changing a workflow setting — which this audit deliberately avoided, per the "modify nothing" instruction. Name/timestamp/active-state metadata only.

20. Target Clean Repo Structure

Current (top-level, relevant portion)

```
ai-job-agent/  
  AI_JOB_AGENT_SYSTEM_AUDIT.md           (root-level historical doc)  
  AI_JOB_AGENT_FOUNDATIONS_IMPLEMENTATION_REPORT.md  
  DATABASE_PLAN.md  
  daily-news-migrations-report.md  
  docs/  
    PRICING.md, PRODUCTION_READINESS.md, job-ingestion-pilot.md, ...  
    *.pdf (architecture reports)  
    job-source-discovery/                 (80+ CSVs, 5 country pilot dirs)  
  n8n-workflows/ (5 workflow .ts/.json pairs)  
  scripts/, src/, supabase/migrations/, tests/
```

Target

```
ai-job-agent/  
  src/, supabase/migrations/, tests/, scripts/ (unchanged – already clean)  
  n8n-workflows/ (unchanged – already the right shape)  
  docs/  
    PRICING.md, PRODUCTION_READINESS.md, ... (living docs, unchanged)  
    *.pdf (unchanged)  
  archive/  
    legacy-company-research/ ← docs/job-source-discovery/pilots/{apify-*,kuwait,qatar,saudi}-expansion  
    legacy-source-intelligence/ ← (reserved – nothing archived yet)  
    legacy-ingestion-experiments/ ← n8n shadow exports, once formally retired  
    legacy-docs/ ← AI_JOB_AGENT_SYSTEM_AUDIT.md, FOUNDATIONS_IMPLEMENTATION_REPORT.md,  
                  DATABASE_PLAN.md, daily-news-migrations-report.md (moved, not deleted)
```

RECOMMENDATION This is a move, not a rewrite: four root-level docs relocate under `docs/archive/legacy-docs/`, and the five Apify/expansion pilot subdirectories relocate under `docs/archive/legacy-company-research/`. `docs/job-source-discovery/*.csv` (the 8 country-level files actually imported into the DB) stay where they are — they are active seed data, not archive material. No source code, migration, or test moves. Not executed in this session.

21. Freeze / Archive / Deprecate Strategy

- **Freeze now** (no new feature work, keep running as-is): Source Intelligence Analyzer (confirmed deactivated — frozen/legacy), Job Ingestion Pilot Orchestrator (frozen — pattern/reference source only, see Parts 18/19), further CSV registry research.
- **Resolved since this report was first drafted**: pausing the live Source Intelligence Analyzer daily schedule — done, confirmed independently; the Pilot Orchestrator's role — resolved as pattern-only, not production foundation.
- **Founder decision still needed**: reconciling CV Analysis Worker v2 node-name drift.
- **Archive (move, not delete) once this report is approved**: the five Apify/expansion pilot subdirectories; the four root-level historical docs.
- **Do not archive yet**: `docs/job-source-discovery/*.csv` country files (still live seed data), any migration, any test, any currently-referenced doc (PRICING.md, PRODUCTION_READINESS.md, job-ingestion-pilot.md, job-ingestion-database-readiness-audit.md, n8n-workflow-change-process.md).
- **Deprecate (stop using, keep for history)**: the company-registry-as-ingestion-gate *behavior* in the Job Ingestion Pilot Orchestrator's "Is Source Approved?" check — the table stays, the gating role goes, and the workflow itself is frozen as pattern reference only.

22. Provider-First Ingestion Validation

Tier	Evidence found	Verdict
A — in-house Greenhouse/Lever/Workable	Fully proven: 4 real sources, real retry/backoff/dedup/idempotency, 20 real rows ingested in local_pilot mode	Patterns stay and get reused in the new clean ingestion layer; extend coverage using the 78 already-verified suitable_public_atl registry rows. The old workflow itself is not the production runner (see Part 19).
B — vetted multi-ATS / Apify career-page extraction	Apify was used only for one-time <i>company discovery</i> (finding which companies/career pages exist), never for recurring <i>job</i> extraction. No evidence this tier works for ongoing ingestion.	Needs a real, small, benchmarked pilot before any commitment — do not assume it generalizes from the discovery pilots.
C1 — Bayt / MENA	No code, no test, no pilot found	Theoretical — needs benchmarking
C2 — JSearch / secondary aggregator	No code, no test, no pilot found	Theoretical — needs coverage + apply-URL verification first, per your own caveat
C3 — Lebanon/pan-Arab supplementary boards	Raw CSV research exists (lebanon.csv , international-remote.csv); no live adapter	Raw material exists, not implemented
D — RSS / free international remote boards	A real, evaluated pilot exists: docs/job-source-discovery/pilots/himalayas/ (api-contract.md, pilot-report.md, normalized-jobs.csv)	Closest to proven of the unimplemented tiers — worth reviewing first
E — AI/browser fallback, schema- validated, discovery-only	Three untracked, undocumented n8n experiments created today (scrape webpages, Google Maps leads, LinkedIn-via-Apify) — none scoped, none proven	Stop — these are ad-hoc probes, not a validated tier. See Part 26.
F — manual / not automatable	62 registry rows already correctly flagged manual_only	Correct as-is; never attempt automation here

RECOMMENDATION No provider tier beyond A should receive implementation time until Tier A's output actually reaches a user as a real match + cover letter. Coverage claims for B/C1/C2/C3/D remain unverified — benchmark before building. The new ingestion layer is built clean; Tier A's proven patterns (retry/backoff, dedup, idempotency, bounded writes, stale-close) are ported into it as reference logic, not by extending the old workflow file.

23. Matching Architecture Confirmation

RECOMMENDATION The proposed shape (hard filter → embedding shortlist → schema-validated LLM rerank → score+reason+missing skills → persisted idempotent match) remains correct and is not replaced.

- **Job-side embeddings:** cache per job, computed once, reused across every user — job descriptions rarely change after ingestion.
- **Profile embedding:** refresh only on a newly *confirmed* CV analysis — this is already exactly what `matches.cv_analysis_id` is designed around ("a newly confirmed analysis is free to produce a fresh match").
- **Same job×user rescoring:** never automatic under an unchanged confirmed analysis — the unique index `matches_user_job_analysis_key` on `(user_id, job_id, cv_analysis_id)` already makes a duplicate score a no-op by construction.
- **Cost control:** hard filter + shortlist keeps the expensive LLM call off every job×user pair — confirmed necessary by Part 14's cost model (rerank is cheap only because it only sees a shortlist, not the whole catalog).
- **Batch strategy:** periodic batch scoring against newly-ingested/updated jobs, not synchronous per-page-load calls.
- **Notification timing:** queue into `notifications` and deliver on a cadence, not instantly per compute — keeps "curated" positioning intact and lets quota accounting (Part 8) happen cleanly at the delivery event.

No evidence found that argues for replacing this with a single large prompt — the shortlist step is what keeps Part 14/15's margins healthy.

24. Safe Transition Plan

Phase	Objective	Touches	Depends on	Expected result	Validation	Rollback
0	Audit + dependency map	None (read-only)	—	This report	Founder approval	N/A
1	Freeze legacy architecture	Live Source Intelligence Analyzer schedule (pause) — DONE, confirmed	Phase 0 approval	No further real rows written under stale logic	Confirmed inactive via n8n search	Re-activate the schedule (no data loss either way)
2	Reconcile n8n + CV Worker drift	CV Analysis Worker v2 node-name diff only — no functional change without separate sign-off	Phase 1	Confirmed parity or a documented, approved fix	Side-by-side node diff, existing <code>tests/workflow/cv-analysis-worker.test.mjs</code>	No live change made until diff is signed off
3	Archive only obvious safe research (Part 20/21)	File moves under <code>docs/archive/</code> — zero code/schema touch	Phase 0 approval	Cleaner repo, zero behavior change	<code>git status</code> shows only renames; full test suite still green	<code>git mv back</code>
4	Build a clean, provider-agnostic ingestion core (normalize/canonicalize/trust-tag/dedupe), reusing the Pilot Orchestrator's retry/backoff/dedup/idempotency/bounded-write/stale-close <i>patterns</i> as reference logic — not extending the workflow itself	New pipeline code, no schema break	Phase 1	A clean core any future fetcher can call	New tests against fixtures reproducing the 20 real rows' shapes	Feature-flag or revert the new code path
5	Build matching	<code>matches</code> worker, embeddings	Phase 4	Real, persisted matches for real users	New DB tests against <code>matches</code>	Worker is additive; disable without touching schema
6	Build paid-launch-required cover-letter generation	<code>cover_letters</code> worker	Phase 5	Real generated + revisable letters	New DB tests against <code>cover_letters</code> , quota enforcement tests at the founder-confirmed final limits (Student 1 revision, Pro 3)	Additive worker, disable independently
7	Build tracking + approval flow	<code>applications</code> worker, outcome-status addition (Part 12)	Phase 6	Approve → send → track works end-to-end, still gated	Reuse existing approval-gate tests; add outcome-state tests	Additive; existing NOT NULL approval gate is untouched throughout
8	Production verification	Full <code>npm test</code> , staging run	Phases 4–7	Every <code>AGENTS.md</code> §22 checklist item passes	Full test suite + manual flow check	Do not deploy until green
9	Archive/deprecate superseded legacy runtime code	Only code proven replaced by Phases 4–7	Phase 8	Final clean state	Re-run full suite after each removal	Everything moved, nothing deleted outright, until confidence is high

25. What Must Not Be Deleted Yet

- Any of the 92 migrations — database history, never deleted per repo convention.
- `docs/job-source-discovery/*.csv` (8 country files) — still the seed source for 593 live `company_sources` rows.
- `docs/PRICING.md`, `PRODUCTION_READINESS.md`, `job-ingestion-pilot.md`, `job-ingestion-database-readiness-audit.md`, `n8n-workflow-change-process.md` — actively cited by migration comments and this very report.
- The four architecture PDFs just checkpointed — the decision record behind Option B.
- All five tracked n8n workflow source pairs — every one is either active, frozen-but-referenced, or a proven pattern source for the new pipeline.
- `scripts/import-company-registry.mjs` — the only re-runnable path from CSV to DB.
- The five Apify/expansion pilot directories — proposed for *archive* (move), never delete; they are the only evidence behind 593 registry rows' provenance.

26. What Must Stop Receiving Engineering Time

- Further CSV registry expansion — 593 candidate sources already vastly exceed 20 ingested jobs; more research CSVs do not move the product forward right now.
- Source Intelligence Analyzer promotion/retry feature growth — finished, frozen, off the critical path.
- Building more custom ATS adapters beyond the proven three — the 78 already-verified `suitable_public_ats` rows are the higher-leverage next step, not a fourth adapter type.
- Company-by-company source discovery as an ingestion *gate* — this is the entire point of Option B.
- Importing n8n templates blindly.
- Full-CV rewrite per match — never promised, not in scope.
- Auto-apply without approval — forbidden by product rule, and already structurally blocked by schema; do not spend time re-deriving what already exists.
- LLM scoring every job×user pair without a shortlist first.
- Cleanup before a dependency audit — this report is that audit; do not cleanup ahead of it again.
- Creating a new project — explicitly rejected (Option B).
- Extending the old Job Ingestion Pilot Orchestrator workflow itself into the production ingestion path — founder-confirmed: reuse its patterns only, build the new layer clean.
- The three shadow n8n experiments created this session (scrape-webpages agent, Google Maps leads, LinkedIn-via-Apify) — zero product tie-in, and the LinkedIn one is expressly forbidden regardless of tier.
- Further Kuwait/Qatar/Saudi expansion research before Lebanon+UAE actually produce one real end-to-end match and cover letter in production.

27. Exact Implementation Order

1. Freeze the live Source Intelligence Analyzer schedule — **done, confirmed**.
2. Small, safe archive of research docs/pilots (Phase 3) — zero behavior change, restores repo legibility immediately.
3. Build the clean, provider-agnostic ingestion core, reusing Tier A's proven patterns as reference logic, not the old workflow itself (Phase 4).
4. Matching (Phase 5).
5. Cover-letter generation — paid-launch requirement, at the founder-confirmed final quotas (Phase 6).
6. Application tracking + approval flow (Phase 7).
7. Production verification (Phase 8).
8. Final archive/deprecate pass on anything Phases 3–7 actually superseded (Phase 9).

28. Exact Next Task

What Batoul should ask Claude Code to do next:

"Implement the deterministic hard-eligibility location filter for job matching — Student restricted to Lebanon, Pro eligible for Lebanon + Saudi Arabia + Qatar + Kuwait + UAE + verified international remote, explicit relocation preference respected, and fail-closed (reject, never guess) whenever a job's location cannot be determined with confidence. Build it as a pure, unit-tested function against the existing `job_preferences / jobs / plans` schema and the 20 real jobs already in the database — no new migration, no new n8n workflow, no LLM calls yet."

RECOMMENDATION This is low risk (pure function, no schema change), measurable (unit tests against known job/preference fixtures), reversible (delete the module, nothing else depended on it yet), directly aligned with the provider-first pipeline's first real stage, and useful immediately toward MVP without requiring any pricing or founder decision first.

Later, deeper sessions in this same conversation refined this into a two-step sequence (deterministic location normalization first, then the eligibility filter on top of it) — see the session's later architecture-sequencing discussion for the current, more granular next-task recommendation. This section is left as originally written since it was not factually wrong, only later superseded by more detailed sequencing.

29. Founder Decisions Required

Decision	Why it can't be made silently	Outcome
Pausing the live Source Intelligence Analyzer's daily schedule	It was an active, live, scheduled workflow currently writing real data	DECIDED — deactivated by the founder; independently confirmed
Student cover-letter revisions 1→2	Pricing/quota change	DECLINED — stays at 1, unchanged
Pro match quota 45→50	Pricing/quota change	DECLINED — stays at 45, unchanged
Pro price (\$18 vs. a future \$22, already rehearsed in docs/PRICING.md 's own example)	Pricing change — recommended to revisit after data, not now	Open — no change made, none proposed for now
Job Ingestion Pilot Orchestrator's role in the new architecture	Whether it becomes the production Tier A runner or stays pattern-reference-only	DECIDED — pattern/reference source only; new ingestion layer built clean
Disposition of the three shadow n8n experiments (scrape agent, Google Maps, LinkedIn-Apify)	Live workflows not authored by this session; deleting/archiving them is a judgment call outside this audit's authority	Open
Whether "AI/TECH NEWS AUTOMATION" (live, active, not in repo) belongs to this project	Ambiguous ownership	DECIDED — confirmed in-scope, intentionally active, powers the website; left untouched
Whether/when to add the user-facing application "outcome status" field (Part 12)	Schema addition	Open
Approving the archive moves in Part 20/21	Repo structure change, even though content is preserved	Open
Confirming cover-letter generation as a hard paid-launch requirement (Part 9/16)	Changes launch scope/sequencing commitment	Open

30. Final Verdict

The branch is clean and exactly where the last two checkpoint commits left it. Architecture Option B is well supported by the evidence found — nothing here argues for a rebuild. Pricing is fundamentally sound and, per explicit founder decision, **unchanged**: the two quota adjustments this report originally proposed (Student revisions, Pro matches) were considered and declined — all plan numbers stay exactly as originally shipped. Pro's price is correctly left unchanged for now, to be revisited with real data rather than changed pre-launch. The repository does not need aggressive deletion — it needs a small, safe archive pass. The one piece of live drift that mattered (the still-scheduled Source Intelligence Analyzer) has been resolved: deactivated and confirmed frozen. The Job Ingestion Pilot Orchestrator's role has also been resolved: it is preserved only as a pattern/reference source (retry/backoff, dedup, idempotency, bounded writes, stale-close, error handling) — the new provider-first ingestion layer is built clean, not as an extension of that workflow. The first implementation step remains small, pure, and reversible: deterministic location normalization followed by the hard-eligibility filter — not ingestion, not matching, not cleanup.

Prepared entirely from read-only inspection: git, local Supabase (read-only counts), live n8n workflow search/details, and the repository's own migrations, tests, and docs. No file was edited, no migration run, no workflow changed, nothing committed or pushed during this analysis or its later correction.